COLLEGE OF LAKE COUNTY

Department of Engineering

EGR 260
Introduction to Circuit Analysis

Lab #4 Superposition

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Submitted to: Dr. Regez

1 Objective

The objective of this experiment is to calculate the load voltage and resistance of a simple resistive circuit using the superposition theorem, simulate power supply and current supply separately and confirm superposition theorem.

2 Equipment

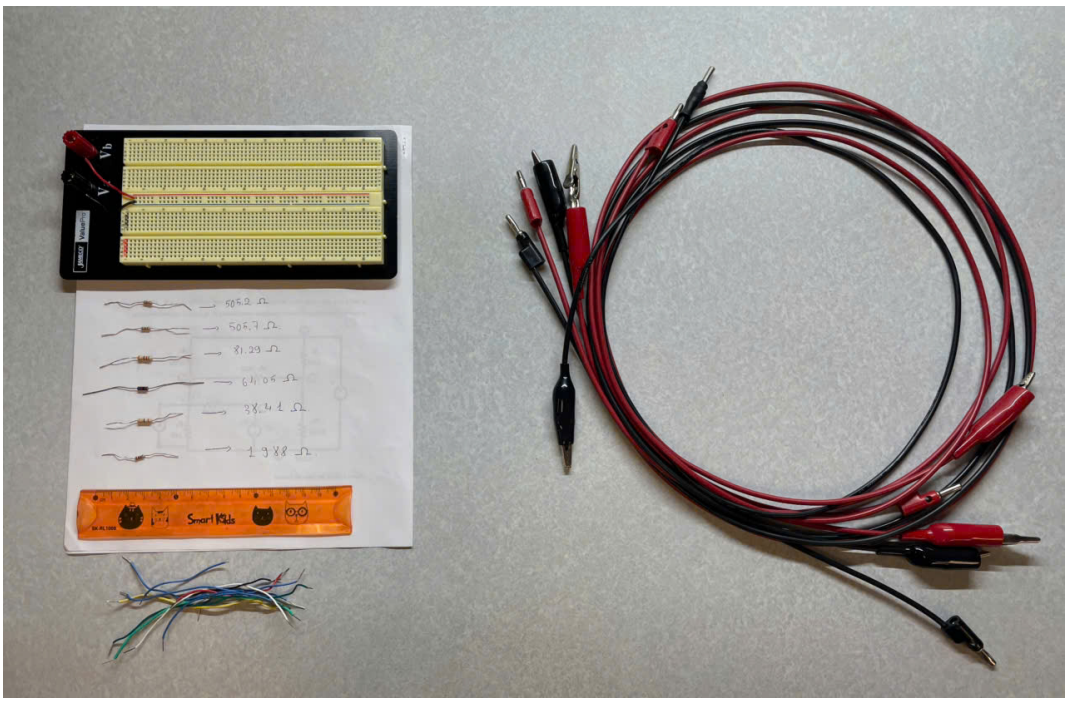

Figure 1: The picture displaying connecting wires, breadboard, male-to-male jumper wires, and six resistors

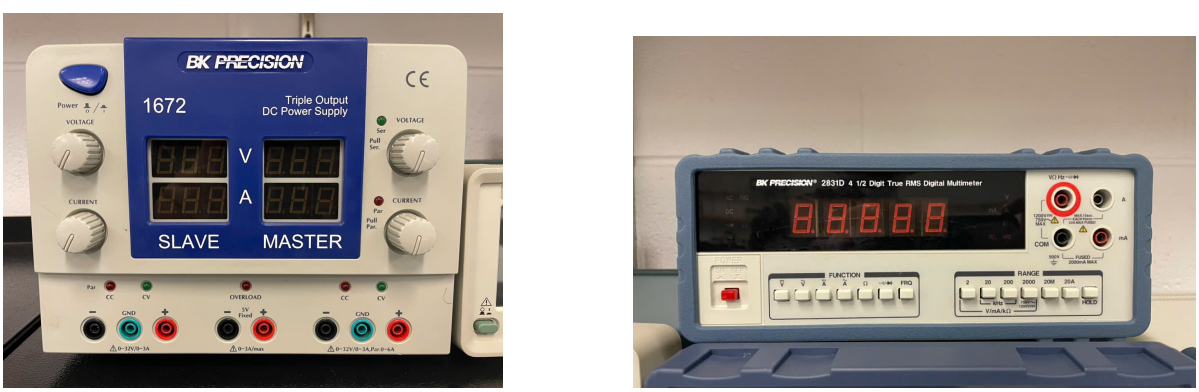

Figure 2: Two pictures displaying the BK Precision DC Power Supply (left) and the BK Precision Digital Multimeter (right) used in this experiment.

3 Experimental Procedure

Figure 3 shows the schematic of the circuit used in part 1 of the experiment.

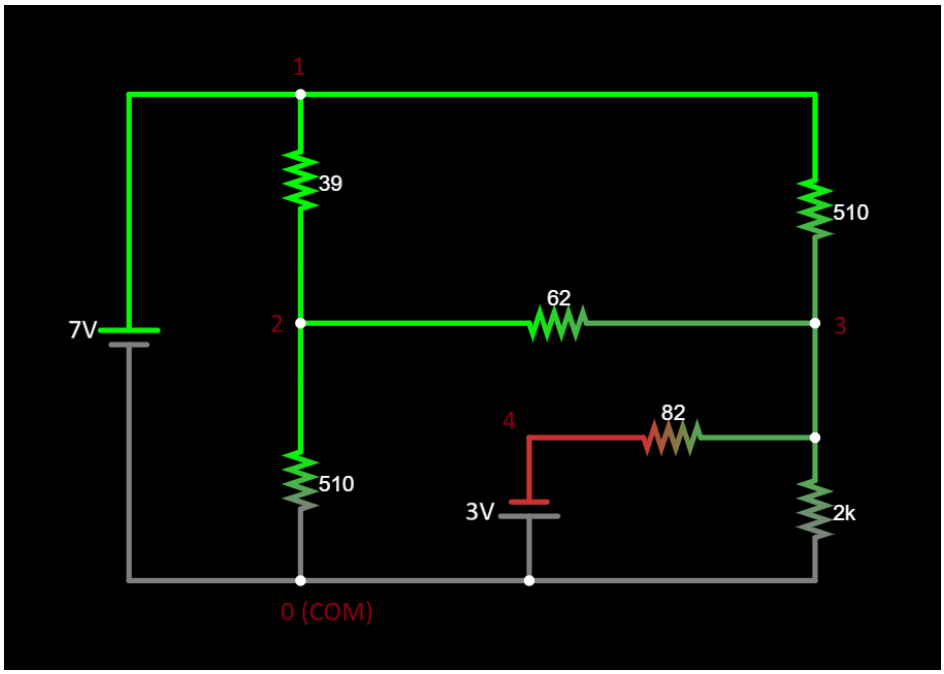

Figure 3: Circuit Schematic

The experimental procedure

1. Gather six resistors, measure, and record the actual resistances shown in Figure 3.
2. Integrate the resistors and jumper wires into a breadboard and construct as shown in Figure 3.
3. Turn off power supply and insert an ammeter between the 62 ohm resistor (R5) and node 3.
4. Turn on both 7V and 3V sources and measure the current through R5
5. Next, record the current due to each power supply acting independently by replacing the 7V supply with a short circuit.
6. Then, record the current between node 3 and R5.
7. Replace the 3V source, restore the 7V source, and repeat step 6.

4 Analysis

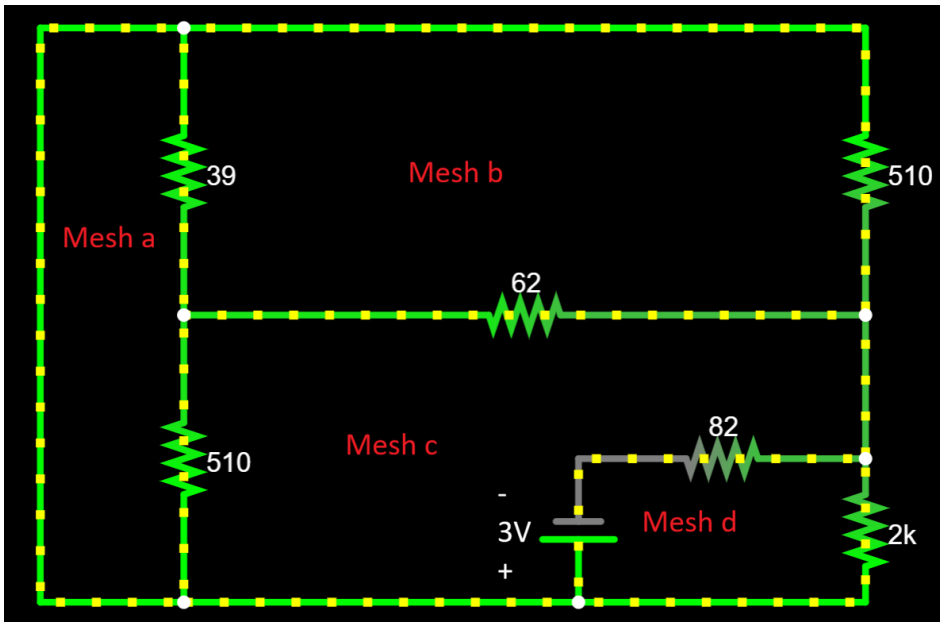

Figure 4: The picture displaying the circuit after shorting the 7V source

Starting with mesh a, use the mesh-current method to find the current through resistor R5:

$$39(i_a - i_b) + 510(i_a - i_c) = 0$$

$$549i_a - 39i_b - 510i_c = 0 \tag{04.01}$$

Mesh-current method with mesh b:

$$39(i_b - i_a) + 510i_b + 62(i_b - i_c) = 0$$

$$-39i_a + 611i_b - 62i_c = 0 \tag{04.02}$$

Similarly with mesh c:

$$62(i_c - i_b) + 82(i_c - i_d) + 510(i_c - i_a) - 3 = 0$$

$$-510i_a - 62i_b + 654i_c - 82i_d = 3 \tag{04.03}$$

Finally mesh d:

$$82(i_d - i_c) + 2000i_d + 3 = 0$$

$$-82i_c + 2082i_d = -3 \tag{04.04}$$

From the equations (04.01), (04.02), (04.03), (04.04), form and solve the matrix:

$$\begin{vmatrix} 549 & -39 & -510 & 0 \\ -39 & 611 & -62 & 0 \\ -510 & -62 & 654 & -82 \\ 0 & 0 & -82 & 2082 \end{vmatrix} \begin{vmatrix} i_a \\ i_b \\ i_c \\ i_d \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 3 \\ -3 \end{vmatrix}$$

After solving the matrix using inverse matrices:

$$i_a = \frac{1308450}{77794709}$$

$$i_b = \frac{224700}{77794709}$$

$$i_c = \frac{1391325}{77794709}$$

$$i_d = \frac{-114597}{155589418}$$

With the calculated mesh currnents, calculate the current through R5:

$$i_{01} = i_c - i_b = \frac{1391325}{77794709} - \frac{224700}{77794709} = 14.996\,\text{mA}$$

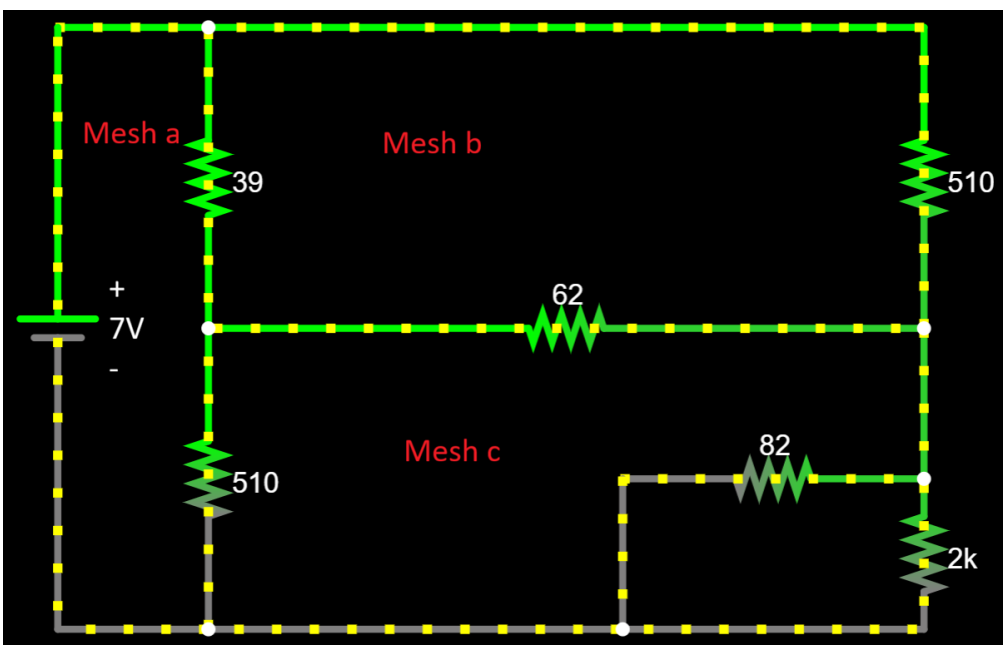

Figure 5: The picture displaying the circuit after resorting the 7V and shorting the 3V source

Similar to the previous part, use mesh-current method to find the current through R5. Beginning with mesh a:

$$39(i_b - i_a) + 510i_b - 7 = 0$$

$$549i_a - 39i_b - 510i_c = 7 \tag{04.05}$$

Mesh-current method around mesh b:

$$39(i_b - i_a) + 510i_b + 62(i_b - i_c) = 0$$

$$-39i_a + 611i_b - 62i_c = 0 \tag{04.06}$$

Eventually around mesh c:

$$62(i_c - i_b) + 510(i_c - i_a) + (82 \parallel 2000)i_c = 0$$

$$-510i_a - 62i_b + 650.77i_c = 0 \tag{04.07}$$

From (04.05), (04.06), (04.07), form a matrix:

$$\begin{vmatrix} 549 & -39 & -510 \\ -39 & 611 & -62 \\ -510 & -62 & 650.77 \end{vmatrix} \begin{vmatrix} i_a \\ i_b \\ i_c \end{vmatrix} = \begin{vmatrix} 7 \\ 0 \\ 0 \end{vmatrix}$$

After solving the matrix:

$$i_a = 0.05123$$

$$i_b = 0.0074155$$

$$i_c = 0.04085$$

From the calculated mesh currents, calculate the current between resistor R5 and node 3:

$$i_{02} = i_c - i_b = 0.04085 - 0.0074155 = 33.438\,\text{mA}$$

According to the superposition theorem, the total current is equal to the sum of i_{01} and i_{02}:

$$i_{total} = i_{01} + i_{02} = 14.996 + 33.438 = 48.434\,\text{mA}$$

5 Data

Resistor	Resistance (Ω)		Current (A)		Voltage drop (V)	
	Measured	Simulated	Measured	Simulated	Measured	Simulated
R1	38.41	39	0.05782	0.05774	2.221	2.252
R2	505.2	510	0.00946	0.00931	4.779	4.748
R3	505.7	510	0.010508	0.010304	5.314	5.255
R4	1,988	2,000	0.000846	0.000873	1.683	1.745
R5	64.05	62	0.048165	0.048435	3.085	3.003
R6	81.29	82	0.057965	0.057866	4.712	4.745

Table 1: Measured and simulated values for the Circuit

Simulated current due to 7V source (mA)	Simulated current due to 3V source (mA)	Simulated current due to both sources (mA)	Measured current due to 7V source (mA)	Measured current due to 3V source (mA)	Measured current due to both sources (mA)
33.438	14.996	48.434	33.81	15.59	49.4

Table 2: Simulated and measured current due to each voltage source and both voltage sources.

6 Graphs

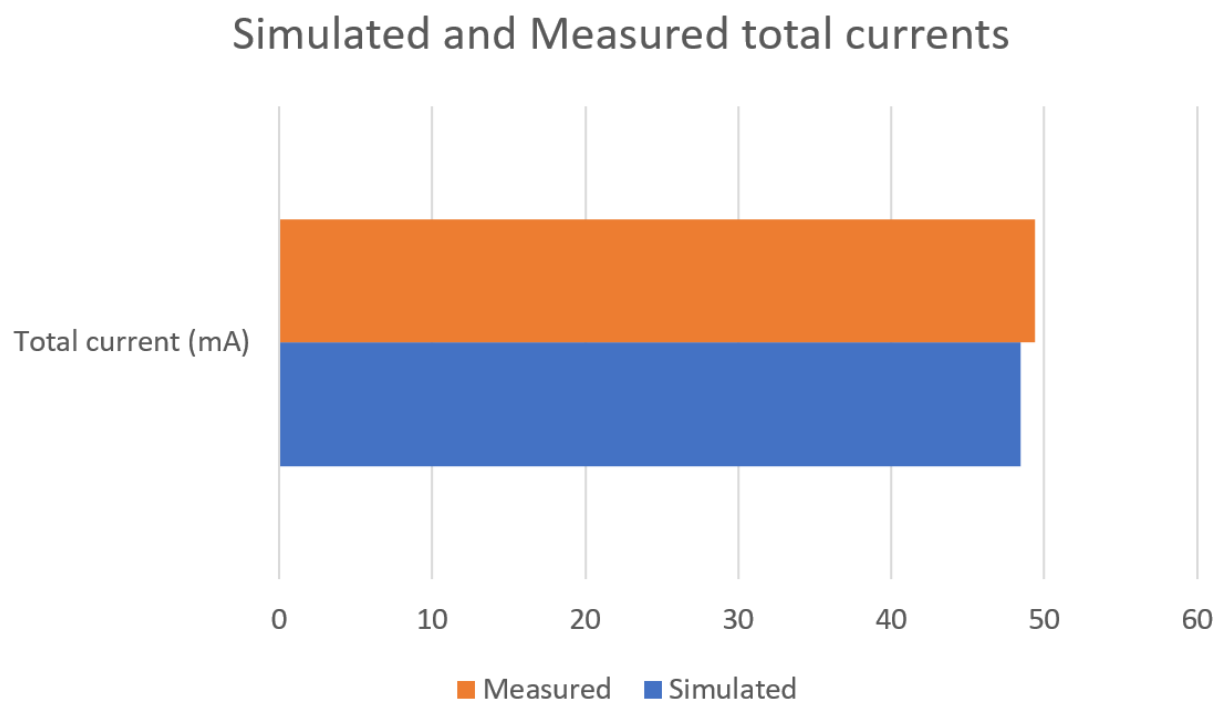

Figure 6: The graph displaying small discrepancy between the measured and simulated total current

7 Discussion and Conclusions

7.1 Discussion

According to superposition theorem, the total current at a given point is the sum of the currents due to each voltage/current source in the circuit. In the analysis part, the circuit was analyzed when voltage source acted individually and the currents due to each sources added up to the simulated total current through resistor R5, successfully confirming the superposition theorem.

Regarding to discrepancy, the percent error stayed relatively small mainly due to difference in the actual and theoretical resistance measurements.

7.2 Conclusion

Overall, the experiment completed the given objective by calculating the current due to each voltage source at a given point, verifying the superposition theorem as the analysis showed the individual currents added up to the total current.